

SOURAV PATTANAİK

National Institute of Science Education and Research (NISER), Jatni, India

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Integrated Master's student in Biology with research interests in structural bioinformatics, molecular simulations, systems biology and computational drug discovery. My work focuses on integrating biological interaction networks with molecular modelling to identify therapeutic targets and design novel therapeutic strategies.

EDUCATION

2021 – 2026	Integrated Master of Science in Biology (Minor degree in Chemistry) National Institute of Science Education and Research (NISER), India CGPA: 8.40 / 10.0 [Rank 16 / 163]
2021	12th Boards (CBSE) DAV School — 94.6% [Top 10 / 139]
2019	10th Boards (ICSE) Chinmaya Vidyalaya (E.M.) — 93.2%

PUBLICATIONS

Rana S., **Pattanaik S.**, Thakur R., Joshi V., Jindal N., Tiwari R. R., & Sahoo G. C. (2026). Computational design and validation of a multi-epitope vaccine candidate against structural proteins of langya virus. Letters in Drug Design & Discovery, 100296. (<https://doi.org/10.1016/j.lidd.2026.100296>) [Accepted]

Pattanaik, S., Konkimalla, V. B. Directional Biological Network (DBN) Analysis Pro: automated target prediction analysis. [Under Preparation]

RESEARCH EXPERIENCE

2024 – 2026	MASTER'S THESIS Supervisor: Dr. V Badireenath Konkimalla, NISER, India <i>"Integrating network-based drug discovery and stimulus-responsive drug carrier for treating testicular cancer through an in silico approach."</i> - Designed a novel dual-stimulus responsive drug carrier using Intrinsically Disordered Proteins(IDPs). - Designed a Gradient Boosting prediction model associated to IDPs with a predictive accuracy of 97.3%. - Developed a computational tool to upgrade static PPI. - Screened for a potential drug molecule among the 17,94,286 molecules in the Pub-Chem Database for binding to the therapeutic target. - Techniques used: Structure Prediction, CG/Atomistic Molecular Dynamics Simulation, Protein-DNA Docking, Network analysis.
May – July 2025	SUMMER PROGRAMME Dr. Anbazhagan Kolandaswamy, RRMCH Bengaluru <i>"Discovery of Biomarkers for Sepsis Disease through comparative analysis of gene expression profiles."</i> - Optimized RNA extraction protocols and processed 52 patient blood samples, while maintaining high-integrity RNA extraction for quantitative profiling of 15 genetic targets(mRNA/miRNA), which were predicted to be differentially expressed using literature and Databases like miRTarBase. - Identified and statistically validated 6 potential biomarkers(P-values < 0.05). - Techniques learnt: qPCR, RT-PCR, Biostatistical analysis, Flow cytometry for immunophenotyping, FlowJo software for analysis, gel electrophoresis.
Jan – Mar 2025	REMOTE COLLABORATION Dr. Karoline Faust, KU Leuven <i>"Beta-testing of a new database associated with Systems Microbiology."</i> - Extracted, structured, and interpreted growth data of 10 different bacterial species. - Contributed to Beta-testing of the new database. - Acknowledged in research pre-print (doi:10.1101/2025.10.13.682118).
May – July 2024	SUMMER INTERNSHIP Dr. Sindhuprava Rana, ICMR-NIREH Bhopal <i>"Multi-Epitope Vaccine design against the structural proteins of Langya Virus."</i> - Used the protein sequence of 5 viral structural proteins to predict different properties like antigenicity, toxicity, etc. - Suggested a novel strategy to include epitopes in the construct while maintaining solubility and biocompatibility. - Published the research project as a co-first author (DOI mentioned above).

- Nov – Dec 2023 WINTER INTERNSHIP** | Dr. Abdur Rehman, Biochemistry lab, NISER
"Introduction to Life Cycle of Tetrahymena."
- Learned confocal microscopy and cell imaging using ImageJ.
- May – July 2023 SUMMER SCHOOL** | Dr. S Mirza, IIT Indore
"Principles of Drug Discovery and Development"
- Focus: Understanding the activity of anti-inflammatory drugs and stages of drug discovery.
- May – July 2022 SUMMER INTERNSHIP** | Dr. C S Purohit, Synthetic Chemistry Lab, NISER
"Multi-step Organic synthesis of 2,6-Bis{[bis(2-pyridinylmethyl)amino]methyl}-4-(1,1-dimethylethyl)phenol by aminomethylation."
- Techniques utilized: Synthesis, TLC, Column Chromatography, NMR.

CONFERENCES AND CONCLAVE

- **Conferences:**
Quarterly Union for Academic Disclosure (QUAD)
IAS-SRFP 36th Mid-Year meeting 2025
InBix 2025.
- Participated in the **Springer Nature India Research Conclave** (2023)

TECHNICAL SKILLS

- **Computational:** Molecular Docking, MD Simulation, Sequence Alignment, Cell imaging, Machine Learning, Biological Network Interpretation, Pharmacophore Modeling, In-silico cloning
- **Wet-lab:** qPCR, RT-PCR, Gel electrophoresis, Plasmid/RNA/DNA isolation, Flow-cytometry, X-ray Crystallography, Protein purification, Cell culture, Confocal Microscopy, ELISA, Synthesis, NMR, Column Chromatography.
- **Additional:** LINUX, LaTeX, Java, Python, AI, Microsoft Office tools, GPU-based Computing, Website and Web tools development.

AWARDS, FELLOWSHIPS AND SCHOLASTIC ACHIEVEMENTS

- **Competitive Exams Qualified:** National Entrance Screening Test 2021 [All India Rank (AIR) 263]; Graduate Aptitude Test in Engineering 2024 [AIR 1173]; Odisha Joint Entrance Examination 2021 [State Rank 3]; Odisha University of Agriculture and Technology 2021 [AIR 81]; National Eligibility cum Entrance Test 2021.
- **Scholarships/Fellowships:** Department of Atomic Energy (DAE)-DISHA Scholarship of 400000 INR(2021-2026); Indian Academy of Sciences-Summer Research Fellowship 30000 INR(May – July 2025).
- **Competitions:** Finalist of BENSPARK Ideathon, IISER Pune [2025]; Excelling ranks in Olympiads (English and Science) [2017-2019]; Won multiple inter-school and intra-college quizzes.

LEADERSHIP AND RESPONSIBILITY

- **Gym Treasurer (2024-2025):** Handled expenditure and bills amounting to almost 50000 INR for NISER Gym Club; Conducted Power-lifting events and served as a coordinator for multiple events.
- **Student Representative:** Member of the Disciplinary Action Committee (DAC) for Students (2025-2026). Contributed actively to policy making and adjudicated cases like substance abuse and harassment to ensure a safe campus environment.

HOBBIES AND EXTRA-CURRICULAR

- **Sports:** Represented NISER in Basketball (IISM 2023) and won inter-batch Basketball tournament (2024). Represented batch in Football (2024) and Badminton (2025) while winning bronze medals.
- **Cultural:** Won multiple dance and slogan writing competitions.
- **Other Interests:** Gym; Reading about novel therapeutic strategies in free time.

REFEREES

Dr. Sindhuprava Rana
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Assoc. Professor, NISER
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Dr. A. Kolandaswamy
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